

1 The 26-Card Disease Model

In this activity, we model the spread of an infectious disease using a deck of playing cards. The model is intentionally simple, but it captures several important features of epidemics: randomness, transmission by contact, and the gradual buildup of immunity in a population.

You will simulate several outbreaks and record how they evolve over time. Even though each group follows the same rules, the results may differ significantly.

1. Work in pairs or small groups. Each group will need one deck of playing cards (no jokers).
Separate the deck into:

- 26 red cards (susceptible individuals),
- 26 black cards (immune individuals).

Place all black cards face up in a pile. This is the *immune pile*.

Hold the 26 red cards face down in your hands. This is the *population pile*.

Throughout the activity:

- red cards represent people who can still become infected,
- black cards represent people who are already immune.

2. Step 0: The initial infection

Remove the top card from the population pile and place it face up on the table. This represents the first infected individual.

Replace that card in the population pile with one black card from the immune pile.

Record the number of new infections at Step 0.

3. Step 1

Shuffle the population pile without looking at the cards.

Draw two cards from the population pile and place them face up on the table.

Interpret the cards as follows:

- each red card represents a new infection,
- each black card represents contact with an immune individual.

Return any black cards to the population pile.

For each red card drawn, replace one red card in the population pile with one black card from the immune pile.

Record the number of new infections at Step 1.

4. Step 2 and beyond

Suppose Step n produces k new infections.

Then Step $n + 1$ proceeds as follows:

- (a) shuffle the population pile,
- (b) draw $2k$ cards from the population pile,
- (c) count how many of those cards are red,
- (d) replace each red card in the population pile with a black card from the immune pile,
- (e) return any black cards drawn to the population pile.

Continue until a step produces no new infections.

5. Record your epidemic in the table below.

Step	Number exposed	New infections
0	1	
1	2	
2		
3		
4		

Continue the table as needed until the epidemic ends.

6. Create a graph of your epidemic.

- horizontal axis: step number,
- vertical axis: number of new infections.

Compare your graph with those of other groups.

7. Answer the following questions about your simulation.

- (a) How many steps did your epidemic last?
- (b) What was the largest number of new infections in a single step?
- (c) Did the epidemic grow steadily, or did it fluctuate?
- (d) At what point did the epidemic begin to shrink?

8. Compare your results with other groups.

- (a) Did all epidemics last the same number of steps?
- (b) Did some epidemics die out quickly?
- (c) Did some grow much larger than others?
- (d) What features did most epidemics have in common?

9. Interpret the structure of the model.

- (a) Why does the population pile always contain 26 cards?
- (b) Why does the epidemic eventually end in this model?
- (c) What real-world feature is represented by replacing red cards with black cards?
- (d) In what ways is this model realistic?
- (e) In what ways is it unrealistic?

10. Explore how changes to the model affect the epidemic.

- (a) What changes if Step 0 begins with two infected individuals instead of one?
- (b) What changes if each infected individual exposes three people instead of two?
- (c) What changes if some individuals begin the activity already immune?

2 From Simulation to a Mathematical Model of an Epidemic

In the previous activity, you simulated the spread of a disease using a deck of cards. Each infected individual exposed two others, and some of those exposures produced new infections while others did not.

In this worksheet, we develop a mathematical model that describes what typically happens in such an epidemic. Instead of tracking one specific outbreak, we study the *expected number* of new infections at each step.

1. Recall your epidemic simulation from the previous activity.
Let I_n represent the number of new infections at Step n .
 - (a) What was the value of I_0 in your simulation?
 - (b) What was the largest value of I_n that occurred?
 - (c) Did your sequence of values increase steadily, fluctuate, or decrease immediately?
2. In the card model, each infected individual exposed two other members of the population.
Suppose there are I_n new infections at Step n .
 - (a) How many total exposures occur at Step $n + 1$?
 - (b) If every exposure produced a new infection, what would I_{n+1} equal?
3. In reality, not every exposure produces a new infection.
Let p represent the probability that a single exposure produces a new infection.
Suppose there are $2I_n$ exposures at Step $n + 1$.
 - (a) On average, how many of those exposures will produce infections?
 - (b) Write a formula for the expected number of new infections at Step $n + 1$.
4. The previous reasoning leads to the approximation

$$I_{n+1} \approx 2pI_n.$$

This is called a *difference equation*. It predicts how the number of infections changes from one step to the next.

- (a) If $p = 0.25$, compute I_1 when $I_0 = 1$.
 - (b) Use your answer to compute I_2 .
 - (c) Continue one more step to compute I_3 .
5. The model

$$I_{n+1} = 2pI_n$$

predicts whether an epidemic grows or shrinks.

- (a) What happens if $2p > 1$?
 - (b) What happens if $2p < 1$?
 - (c) What happens if $2p = 1$?
6. The condition

$$2p = 1$$

represents a boundary between epidemic growth and epidemic decline.

Solve this equation for p .

This value of p is called a *threshold value*.

- (a) What happens when p is larger than this value?
 - (b) What happens when p is smaller than this value?
7. In the card model, each infected individual exposed two others.
Suppose instead that each infected individual exposed c others.
 - (a) Write a new difference equation for I_{n+1} .
 - (b) What condition determines whether the epidemic grows?

8. The quantity

$$R = cp$$

represents the expected number of new infections caused by one infected individual.

- (a) Explain what $R > 1$ means in this model.
- (b) Explain what $R < 1$ means in this model.
- (c) Why might public-health officials want to reduce the value of R ?

9. Return to the original card simulation.

- (a) Early in the simulation, was the fraction of red cards in the population pile large or small?
- (b) Later in the simulation, did this fraction increase or decrease?
- (c) How did this change affect the number of new infections at each step?
- (d) Why does this explain why epidemics eventually end in the card model?

10. So far, we have studied how the number of new infections changes from one step to the next.

In the next activity, we will track three groups in a population simultaneously:

- susceptible individuals,
- infected individuals,
- recovered individuals.

This leads to one of the most important models in mathematical epidemiology.

3 Building the SIR Model

In previous activities, we modeled how infections spread step by step using a card simulation and a difference equation for the number of new infections. In this worksheet, we refine that approach by tracking three groups in a population simultaneously:

- susceptible individuals,
- infected individuals,
- recovered individuals.

This leads to one of the most important models in mathematical epidemiology: the SIR model.

1. Suppose a disease spreads through a population of fixed size N .

At Step n , let:

- S_n be the number of susceptible individuals,
- I_n be the number of infected individuals,
- R_n be the number of recovered individuals.

What equation must always be true relating S_n , I_n , and R_n ?

2. To model how infections occur, we make the following assumption:

The number of new infections during Step n is proportional to both the number of susceptible individuals and the number of infected individuals.

- (a) Why should the number of new infections increase if S_n increases?
- (b) Why should the number of new infections increase if I_n increases?
- (c) Explain why it is reasonable to model new infections using a product involving S_n and I_n .

3. Suppose new infections occur at rate

$$\beta S_n I_n,$$

where β is a constant describing how easily the disease spreads.

- (a) What does a larger value of β represent?
- (b) What does a smaller value of β represent?

4. Suppose infected individuals recover at rate

$$\gamma I_n,$$

where γ represents the fraction of infected individuals who recover during one step.

- (a) What does a larger value of γ represent?
- (b) What does a smaller value of γ represent?

5. Susceptible individuals become infected, so the susceptible population decreases each step.

Using the infection term $\beta S_n I_n$, write an equation for S_{n+1} .

6. The infected population changes in two ways:

- it increases due to new infections,
- it decreases due to recoveries.

Using the infection term $\beta S_n I_n$ and the recovery term γI_n , write an equation for I_{n+1} .

7. Recovered individuals come only from the infected population.

Using the recovery term γI_n , write an equation for R_{n+1} .

8. Combine your previous answers to write the complete system:

$$S_{n+1} =$$

$$I_{n+1} =$$

$$R_{n+1} =$$

This system is called the discrete SIR model.

9. Show that the total population remains constant from one step to the next.

That is, verify that

$$S_{n+1} + I_{n+1} + R_{n+1} = S_n + I_n + R_n.$$

10. Suppose the disease is just beginning to spread, so that almost everyone is susceptible.

That means we can approximate

$$S_n \approx N.$$

Use the equation for I_{n+1} to determine when the number of infected individuals increases from one step to the next.

Your answer should produce a condition involving β , γ , and N .

11. The quantity

$$R_0 = \frac{\beta N}{\gamma}$$

is called the *basic reproduction number*.

(a) What does $R_0 > 1$ mean?

(b) What does $R_0 < 1$ mean?

(c) How does this relate to the threshold condition $cp > 1$ from the previous worksheet?

12. The discrete SIR model tracks changes step by step. A closely related model instead tracks how these quantities change continuously over time.

In that setting, the model becomes

$$S'(t) = -\beta S(t)I(t)$$

$$I'(t) = \beta S(t)I(t) - \gamma I(t)$$

$$R'(t) = \gamma I(t).$$

In later work, we will use this continuous model to predict epidemic peaks, vaccination thresholds, and long-term outcomes.

4 Compartment Models and Estimating Parameters from Data

In earlier worksheets, we built the SIR model by tracking three groups: susceptible, infected, and recovered individuals. In this worksheet, we extend this idea to more general compartment models and learn how to estimate model parameters using epidemic data.

1. A *compartment model* divides a population into groups that represent disease status.
Examples include:

- susceptible individuals,
- infected individuals,
- recovered individuals,
- exposed individuals who are infected but not yet infectious.

Explain why it is useful to divide a population into compartments when modeling an epidemic.

2. Recall the discrete SIR model:

$$\begin{aligned}S_{n+1} &= S_n - \beta S_n I_n \\I_{n+1} &= I_n + \beta S_n I_n - \gamma I_n \\R_{n+1} &= R_n + \gamma I_n\end{aligned}$$

Identify which term represents:

- (a) new infections,
 - (b) recoveries.
3. Some diseases include an incubation period during which individuals are infected but not yet infectious.
To model this, we introduce a new compartment:
 E_n = number of exposed individuals.
Individuals move through compartments in this order:
susceptible $\rightarrow$ exposed $\rightarrow$ infected $\rightarrow$ recovered
Explain why adding an exposed compartment may improve the realism of a model.
 4. Suppose exposed individuals become infectious at rate

$$\sigma E_n.$$

Write equations describing how the exposed compartment changes from Step n to Step $n + 1$.

5. Combine the infection term $\beta S_n I_n$, the progression term σE_n , and the recovery term γI_n to write a discrete SEIR model.

$$S_{n+1} =$$

$$E_{n+1} =$$

$$I_{n+1} =$$

$$R_{n+1} =$$

6. The parameters β , σ , and γ describe how quickly individuals move between compartments.
 - (a) What does β measure?
 - (b) What does σ measure?
 - (c) What does γ measure?
7. Early in an epidemic, most individuals are susceptible. This allows us to approximate

$$S_n \approx N.$$

Using this approximation, simplify the equation for I_{n+1} .

8. Suppose infection counts during the first few days of an outbreak are:

Day	Infected
0	5
1	8
2	13
3	21

(a) Compute the ratios I_1/I_0 , I_2/I_1 , and I_3/I_2 .

(b) Estimate the growth factor of the epidemic.

9. Suppose the early-stage approximation

$$I_{n+1} \approx (1 + \beta N - \gamma)I_n$$

holds.

Use your estimated growth factor to estimate the value of

$$\beta N - \gamma.$$

10. Recall that the basic reproduction number is

$$R_0 = \frac{\beta N}{\gamma}.$$

Use your estimate from the previous exercise to approximate R_0 .

11. Suppose a disease has estimated value $R_0 = 2.4$.

(a) What does this number mean?

(b) Would you expect infections to increase or decrease initially?

(c) What kinds of interventions might reduce R_0 ?

12. Compartment models simplify many aspects of disease spread.

List at least three assumptions made by the SIR or SEIR models that may not hold in real populations.

13. Realistic epidemic models often include additional compartments such as:

- vaccinated individuals,
- hospitalized individuals,
- asymptomatic individuals.

Describe one additional compartment you would add to improve realism, and explain how individuals would move into and out of it.